

PROMPT 1

I am working on a Python Pygame project called Galaga Repair Lab. I will provide you with my game.py source code.

Your first task is to analyze the entire codebase and help me understand how the game works.

Focus on the following:

1. Explain the overall game architecture and the purpose of its major functions and classes.
2. Explain how enemy spawning, Bézier entry paths, enemy diving, shooting, and collision detection work.
3. Identify the deliberate bug in the bezier() function by comparing it with the standard cubic Bézier formula.
4. Explain exactly why the incorrect exponents cause the enemies to bulge or overshoot during movement, even though the endpoints remain correct.
5. Do not modify the code yet. First, explain the problem in simple terms and show me the incorrect and correct mathematical expressions.

I want to understand the bug before fixing it, so explain it as if I am a beginner in Python and game development.

PROMPT 2

Now let's fix and improve my Galaga Repair Lab project.

Based on your previous analysis, help me implement the following four tasks:

TASK 1: Fix the Bézier curve bug.

- Correct the exponents in the two middle terms of the bezier() function.
- Preserve the standard cubic Bézier formula.
- Do not unnecessarily modify other game mechanics.

TASK 2: Implement enemy_tint(kind).

- Support boss, red, and blue enemies.
- Make enemy colours change based on the current wave.
- Ensure the returned value is a valid RGB tuple or None.

TASK 3: Implement on_wave_start(wave).

- Display a WAVE X banner whenever a new wave starts.
- Keep the banner visible for approximately one second.

- Ensure the banner does not interfere with gameplay.

TASK 4: Implement shield_charges(wave).

- Grant one shield charge every three waves.
- Ensure the existing hit_player() shield logic works correctly.

IMPORTANT:

- Work with my existing code instead of rewriting the entire game.
- Explain every modification and why it is necessary.
- Give me the exact functions or code sections I need to replace.
- Maintain compatibility with the existing game architecture.
- Avoid introducing new bugs.

Let's implement these changes step by step. Do not skip explanations.

PROMPT 3

I have implemented the changes in my Galaga Repair Lab project. Now act as a game-testing and debugging assistant.

Help me verify whether the game meets all the assignment requirements.

Test and review the following:

1. Bézier movement:
 - Enemies should enter smoothly.
 - They should pass through the correct start and end points.
 - There should be no unnatural bulging or overshooting.
2. Enemy behaviour:
 - Enemies should enter formation, sway, dive, and shoot.
 - Boss enemies should require two hits.
 - Each bullet should destroy at most one enemy.
3. New features:
 - Enemy colours should change according to wave progression.
 - The WAVE X banner should appear for approximately one second.
 - Shield charges should be granted every three waves and consumed correctly.
4. Game mechanics:
 - Player movement and shooting should work.
 - Extra ships should be awarded according to the existing kill logic.
 - Losing all ships should reduce lives.
 - The game should end when lives reach zero.
 - New waves should spawn correctly.
5. Check for:

- Runtime errors.
- Incorrect variable initialization.
- Timing issues.
- Collision detection problems.
- Any unintended changes to the original game mechanics.

Give me a final checklist of passed and failed tests. If you find problems, explain their cause and provide the smallest possible correction.

Finally, tell me exactly how to demonstrate the before-and-after gameplay for my 10-second submission videos.